TRIARCH
CONSULTING LLC

RELATIONSHIP ENGINE · GOVERNANCE & COMPLIANCE

DOMAIN DEMO PACKET · EXPLAINER PLAYBOOK V0.1

Governed Knowledge in a Hard Domain

How a relationship & governance engine fits around cancer research — without ever practicing medicine.

Corbin Kruczek · Triarch Consulting LLC · June 2026 · Illustrative demo

READ THIS FIRST — WHAT THIS IS AND IS NOT

Triarch is not a medical provider, and nothing here is medical advice. We do not diagnose, treat, recommend treatment, or stand between a patient and a clinician. We are a **governance and provenance layer for information** — the plumbing that tracks what is corroborated, by what evidence, and who is allowed to rely on it.

This packet uses oncology only as a **hard-case illustration** of how that governance layer behaves under pressure. Every clinical decision remains entirely with licensed professionals. If you are a patient: talk to your doctor, not this document.

THE ONE-BOX VERSION

Cancer research already produces more findings than any human or institution can track, and a chronic problem is knowing *which findings are trustworthy, how current they are, and on what evidence.* Triarch doesn't do the science. We do the **governance around the science**: a slim, auditable kernel that holds a body of *corroborated* knowledge, tracks the evidence behind every claim, flags what's unverified rather than guessing, and routes questions to the right specialist resource — while keeping sensitive data local and every output traceable. The disease is the doctors' problem. The *trustworthiness and provenance of the information* is ours.

1 · The Metaphor: A Library That Refuses to Lie

Imagine a research library where every book carries a live, honest label.

In an ordinary library, a confident-looking book and a debunked one sit on the shelf looking identical. You can't tell from the spine which claims have held up and which were quietly overturned. In medical research this is a real and expensive problem — findings fail to reproduce, get superseded, or get over-stated, and the bad ones keep circulating because nothing *marks* them.

Triarch's role is the **librarian who keeps every label honest**. Each claim on the shelf carries: *how strongly it's corroborated, what evidence supports it, when it was last checked, and whether it's still standing or has been demoted.* The librarian doesn't perform surgery or invent cures — they make sure that when a qualified person reaches for a claim, they know *exactly* how much weight it can bear.

WHERE THE METAPHOR BREAKS (WE FLAG IT)

A library is static; corroborated science is not. Our "labels" must update as evidence changes, and a claim that's solid today can be **demoted** tomorrow. So it's less a library than a *living ledger* — which is exactly the discipline Triarch is built on: nothing is ever "verified" and final; things are "corroborated" and stay under watch.

2 · Where the Industry Actually Is (2025–2026)

INDUSTRY CLAIMS IN THIS SECTION CURRENT AS OF: 18 JUNE 2026

The science is moving fast and in a specific direction — and that direction is what creates the governance need.

Three things are true about oncology right now, all well-documented in the current literature:

- **AI is accelerating drug discovery and molecular design.** Machine-learning systems now screen enormous combination spaces — one 2025 effort modeled 1.6 million drug-combination predictions from a few hundred real experiments, validating 307 synergistic combinations[1] — and reviews document AI-designed molecules entering trials in compressed timelines[2].
- **Treatment is becoming adaptive and personalized.** The field is moving away from one-size, max-dose approaches toward *adaptive therapy* — reading a tumor's evolution (e.g. via circulating tumor DNA) and adjusting over time, because tumors are heterogeneous, evolving adversaries rather than fixed targets[2,3].
- **"Digital twins" of disease are an active research concept.** Layered, patient-specific models — general biology, then population strata, then the individual — are being explored as "global learning health and disease models"[4]. The structure is exactly a reference-frame stack: shared priors at the base, specificity at the edge.

THE HONEST GAP THE INDUSTRY OPENLY ACKNOWLEDGES

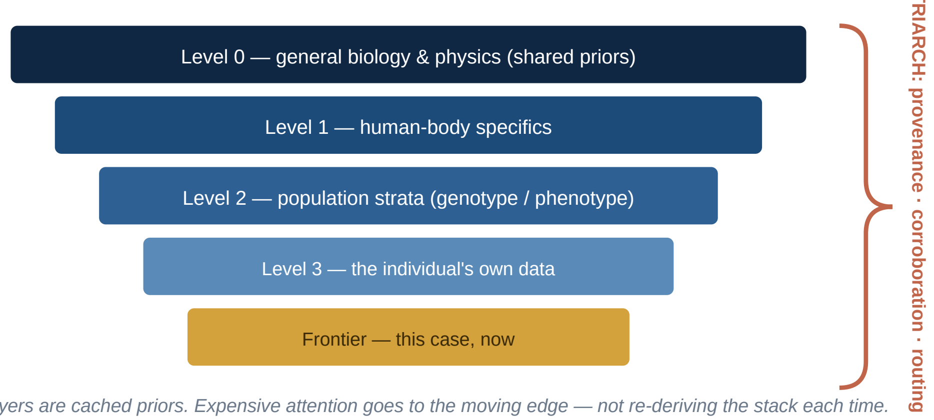
More findings means more *unvalidated* findings. The literature repeatedly names the same constraints: results that don't reproduce, models that need "broader validation, mechanistic understanding, and regulatory alignment"[5], and the difficulty of **knowing which knowledge to trust and act on**. The bottleneck is shifting from *generating* knowledge to *governing* it. That gap is not a medical problem — it's an information-trust problem.

3 · The Triarch Play (Strictly: the Governance Layer)

We sit beside the research, never in the clinic. Our job is provenance, corroboration status, and safe routing.

Picture the layered "reference frame" the field is already reaching for, and notice that Triarch's contribution is the **connective governance** across the layers — not the biology inside them:

A LAYERED REFERENCE FRAME (industry concept) + TRIARCH GOVERNANCE (our role)



Lower layers are cached priors. Expensive attention goes to the moving edge — not re-deriving the stack each time.

Triarch is the vertical governance spine, not the horizontal biology. We track trust and route; clinicians and researchers own the content.

What Triarch does	How it shows up in this domain
Corroboration status, not verdicts	Every claim carries a live status (corroborated / under watch / demoted) with its evidence chain. We never assert a finding is "true and final" — we track how well it has held up.
Flag, don't fabricate	If the evidence for a claim isn't documented, the system says *"not documented,"* never invents support. Unknown stays unknown.
Cache the base, compute the edge	Stable lower-layer knowledge is treated as cached priors; effort concentrates on what's novel at the frontier — efficient and non-redundant.
Data stays local; proofs travel	Sensitive data stays where it belongs. What leaves is a verifiable proof of a result, not the underlying records — provable to an overseer without exposing the data.
Route to the right authority	Questions are dispatched to the appropriate qualified resource. Triarch connects and governs; it does not answer clinical questions itself.

HOW WE GOT HERE (THE HONEST REASONING TRAIL)

This packet deliberately stops at the *governance layer*. We are not claiming to model tumors, design drugs, or compute cures — those are open scientific problems requiring measurement and expertise we don't possess and won't pretend to. What ports cleanly from Triarch's other domains (manufacturing compliance, regulatory workflows) is the **discipline of governed, corroborated, provenance-tracked knowledge with safe routing**. That discipline is domain-agnostic. The biology is not ours; the trustworthiness plumbing is.

4 · Why This Is the Same Pattern Everywhere

The reason this demo matters isn't oncology — it's that the *shape* is identical to every other Triarch domain. A slim governance kernel, a layered set of cached priors, corroboration-not-verification, flag-don't-fabricate, data-local-proofs-out, and routing to the right authority. Swap "tumor research" for "regulatory filings" or "manufacturing quality records" and the kernel is unchanged — only the content of the layers differs. (See the companion compliance demo for the same pattern in a domain with no medical exposure at all.)

References & Sources

ALL SOURCES ACCESSED & VERIFIED: 18 JUNE 2026

1. Narykov O., et al. “AI-driven discovery of synergistic drug combinations against pancreatic cancer.” *Nature Communications* **16**, Article 4020 (2025). nature.com/articles/s41467-025-56818-6 · see also NCATS announcement, ncats.nih.gov
2. “Artificial intelligence in oncology drug development and management: a precision medicine perspective.” *Frontiers in Oncology* **15** (2025). doi.org/10.3389/fonc.2025.1609827
3. “AI-Driven Advances in Precision Oncology: Toward Optimizing Cancer Diagnostics and Personalized Treatment.” *AI* **7**(1):11 (Jan 2026). doi.org/10.3390/ai7010011
4. Li X., Loscalzo J., et al. “Digital twins as global learning health and disease models for preventive and personalized medicine.” *Genome Medicine* **17**:11 (2025). (cited in BJC Reports, nature.com/articles/s44276-026-00221-1)
5. “Precision oncology in the age of AI: lessons from AI-driven drug discovery and clinical translation.” *BJC Reports* (2026). nature.com/articles/s44276-026-00221-1

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About this document. Authored by Corbin Kruczek, Triarch Consulting LLC, as an illustrative demonstration of the Triarch governance architecture. **It is not medical advice and not a medical device or service.** Triarch does not diagnose, treat, or recommend treatment, and does not replace any licensed clinician. Oncology is used here solely as an illustrative hard-case domain. Industry characterizations reflect publicly available literature as of mid-2026 and are general, not exhaustive. Any real deployment in a healthcare context would require